

Journal Preparations

IB 514: Scientific Writing

Author Guidelines

Read them! They can be overwhelming, but focus on the essentials.

Things I ignore:

- Font type and size, margin requirements
- Journal-requested citation format

Things I don't ignore:

- Word/page limits (*incl. title, abstract, main text*)
- Line spacing (*double*) & page numbering
- Figure placement (*inline unless specified*) and formats (*pdf, tiff, eps*)
- Cover page and/or end-of-manuscript requirements
 - Author Contribution Statement
 - Open Science (Data & Code Availability) Statement

Check:

- Journal's double-blind requirements
- Citation completeness and consistency
- Triple-read for spelling and punctuation

Author Contributions Statement

Purpose

- Distinguish **authorship** from **acknowledgment**
- Increase transparency and accountability
- Expose *ghost authorship* and *gift authorship*
- Increasingly indexed by databases (e.g., OpenAlex)

The CRediT Taxonomy (14 Roles) credit.niso.org

Conceptualization	Methodology
Data Curation	Project Administration
Formal Analysis	Resources
Funding Acquisition	Software
Investigation	Supervision
Validation	Visualization
Writing — Original Draft	Writing — Review & Editing

Author Contributions Statement

Common Formats

1 Prose (narrative)

“J.S. conceived the study and led data collection. M.R. performed statistical analysis. . .”

2 Role-tagged list

Conceptualization: J.S., M.R.

Formal Analysis: M.R.

Writing—Original Draft: J.S.

Writing—Review & Editing: All authors

Imperative: Agree on (a draft) contributions statement *before writing starts*, not retroactively. Disagreements are uncomfortable, potentially costly, and typically avoidable.¹

¹See *Collaboration* class.

Page Charges & Publication Fees

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- **Article Processing Charges (APCs):** \$500–\$12,000+ depending on journal/Open Access (OA) model
- **Page charges:** \$0–\$200 per printed page for subscription journals
- **Color figure fees:** \$300–\$1,000+ per figure; often free online-only

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Who Pays — and How to Find Out

- **Ask your advisor first:** most PIs budget for publication costs through grants; this is a normal conversation to have
- **Institutional waivers:** many universities (but not OSU!) have OA publishing funds; check with library
- **Funder mandates:** NIH, Wellcome Trust, and others *require* OA and may cover fees directly
- **Request a waiver:** most journals offer hardship waivers for authors without funding — ask before assuming you must pay

Bottom line: You shouldn't be expected to pay out of pocket. Clarify funding *before* submission, not after acceptance.

Pre-Submission Review by Friendly Reviewers

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Selecting Reviewers

- Avoid/Seek reviewers with strong prior positions on your findings
- Prioritize people who will give *honest* feedback, not reassurance
- Consider who might make a better journal reviewer

Briefing Reviewers

- State the target journal and audience
- Set a realistic deadline (2-3 weeks is standard)
- Clarify what kind of feedback is useful: structure, argument, clarity, or all three
- Identify specific concerns you want addressed

Thank them by name in the Acknowledgments.

Consider mentioning in Cover Letter.

Friendly review is *not* a substitute for rigorous self-revision. Send out only when you believe the manuscript is essentially complete.

Suggesting Potential Reviewers

How to Identify

- Researchers who publish directly on topic or methods
- Authors cited in manuscript (but not exclusively)
- Individuals with complementary expertise (theory, methods, system)

Who to Avoid

- Current or recent collaborators (typically last 3–5 years)
- Members of your institution or close professional network
- Advisors, committee members, or mentors

Strategic Considerations

- Aim for 4–8 suggested reviewers
- Ensure diversity (perspectives, institutions, career stage, gender)
- Avoid suggesting only highly favorable reviewers

Opposed Reviewers

- Allowed by some journals but use very sparingly and justify!²

²Consider use as "*friendly*" reviewer

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Explore your target journal's submission system!

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Type	Who Pays	Description
Gold	Author / funder	Published OA ³ immediately by the journal; APC ⁴
Green	Nobody	Author self-archives a preprint or accepted manuscript in a repository (e.g., biorXiv)
Diamond	Nobody	Community- or institution-funded; no fees for authors or readers
Hybrid	Author	Subscription journal offers OA for individual articles upon APC payment
Bronze	Nobody	Freely readable on the publisher site <i>without</i> open licence; access can be revoked

Note: Green OA costs nothing and is compatible with most journals. Check policies at <https://openpolicyfinder.jisc.ac.uk>.

³OA - Open Access

⁴APC - Article Processing Charges (can be substantial)

Data & Code Sharing

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- Increasingly required by journals and funders
- Can be cited - Digital Object Identifier (DOI)
- Common repositories:
 - Data: Zenodo, FigShare, Data One
 - Code: GitHub, Zenodo
 - Supplementary materials (*no longer considered enough*)
- Best practices:
 - Include comprehensive documentation and README files
 - Provide appropriate licensing

Take *IB 516: Analytical Workflows!*

FAIR Data Principles

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FAIR defines standards to maximise reuse and reproducibility.

F — Findable Data are assigned persistent identifiers (e.g., DOIs) and described with rich, searchable metadata.

A — Accessible Data and metadata are retrievable via open, standardised protocols; access conditions are clearly stated.

I — Interoperable Data use common formats, vocabularies, and ontologies to enable integration with other datasets and tools.

R — Reusable Data carry explicit licences (e.g., CC BY) and sufficient provenance information for others to build on them.

Full guidelines at go-fair.org/fair-principles.

Note: FAIR does not require data to be *open* — sensitive data can be FAIR with controlled access.

Preprint Repositories

Consider posting your ms to a preprint server before or during journal submission.

Advantages

- **Speed:** findings are public immediately, months before formal publication
- **Priority:** timestamped record of contribution
- **Visibility:** indexed by Google Scholar, etc.
- **Feedback:** community comments before peer review
- **Open access:** freely available
- **Funder compliance:** satisfies some open science mandates immediately

Disadvantages

- **Journal policies:** some journals discourage or prohibit
- **No peer review:** errors and weak claims are public
- **Perception:** some fields and hiring committees do not credit preprints
- **Retraction gap:** flawed preprints may persist even after journal rejection
- **Media risk:** premature press coverage before validation

Preprint Repositories

Examples:

bioRxiv Life sciences; operated by Cold Spring Harbor Laboratory; the dominant biology preprint server since 2013.

EcoEvoRxiv Ecology and evolutionary biology; community-governed; supports registered reports and post-publication versions.

arXiv Physics, mathematics, computer science, quantitative biology; the oldest major preprint server (est. 1991); moderated by field endorses.

Zenodo General-purpose; accepts preprints, datasets, code, and presentations; assigns DOIs; widely used for open science compliance.

Authorea Collaborative writing and preprint platform; supports live figures, data embedding, and inline commenting; blurs the boundary between manuscript and publication.

Peer Community In... (PCI)

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PeerCommunityIn.org

Goals

- Transparent peer review *before* (or *instead of*) journal submission
- Eliminate fees for authors and readers
- Reduce dependence on commercial publishers

How It Works

- Authors submit preprint (e.g., to bioRxiv), then request PCI review
- Volunteer recommender (acting editor) solicits and coordinates reviews
- Accepted preprints receive a **PCI Recommendation** — citable endorsement
- PCI-friendly journals accept recommended preprints, or regular review

PCI Animal Science

PCI Archaeology

PCI Ecology

PCI Ecotox Env Chem

PCI Evol Biol

PCI Forest Wood Sci

PCI Genomics

PCI Health Mov Sci

PCI Infections

PCI Math Comp Biol

PCI Microbiology

PCI Network Science

PCI Neuroscience

PCI Nutrition

PCI Org Studies

PCI Paleontology

PCI Plants

PCI Psychology

PCI Registered Reports

PCI Statistics and

Machine Learning

PCI Zoology

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Preprint Repositories

- Final formatting is important, but don't obsess over minor details
- Author contributions should be agreed upon early and transparently
- Clarify publication fees and funding sources before submission
- Pre-submission review by trusted colleagues can improve your manuscript
- Suggesting appropriate reviewers can increase chances of acceptance
- Open science practices enhance transparency and impact, but consider costs and requirements
- Preprint repositories offer rapid dissemination but check journal policies